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# HACKATHON 2026

*Idea Submission Template / Code. Create. Innovate.*

|                         |                                         |
|-------------------------|-----------------------------------------|
| Team Name               | Team Raksha                             |
| Problem Statement No.   | 2                                       |
| Problem Statement Title | Improving Safety at Large Public Events |
| Team Leader Name        | Shounak Joshi                           |

# THE PROBLEM

## Guards watch screens. Danger builds unseen.

A control room watches dozens of flat camera feeds. A guard spots trouble only after the crowd stops moving. By then, crushing pressure is already passing body to body. Stillness is not the warning. It is the accident.

### Hathras 2024

121 dead. 80k permit, 250k crowd. Exits blocked.

### Mumbai 2017

23 dead on a station footbridge. Rain, rush hour, no metering.

### 7 per m2

A crowd turns fluid. Shockwaves throw people metres. (Fruin)

### Why stillness lies

A stopped crowd is not a warning. It is the accident. Pressure passes body to body long before feet stop. That is why cameras must read force, not faces.

## Kumbh, 29 January 2025



### Months before

2,751 cameras and AI alerts go live.



### 01:00

Barricades jumped at the Sangam nose. Crores press forward.



### After

30 dead officially. The alarm watched. Nobody acted.



Kumbh mela ground at night, Jan 2025. Photo: Wikimedia Commons.

*The system saw it coming. No gate opened.*

**Danger is invisible pressure, not visible stillness.**

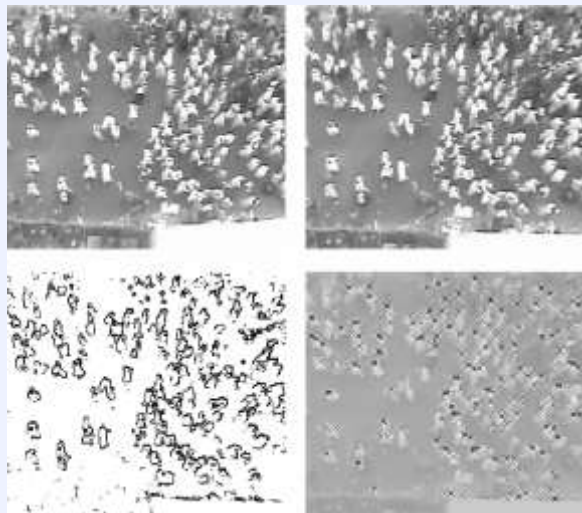
Measure pressure, warn early, open gates.

# HOW IT WORKS



In plain words: map the ground right, count the crowd as heat, read its currents, warn on pressure.

## What the machine sees



Head-density surface: each peak is one head, the sum is the count. (Johansson 2008).



Flow field on our station clip (DIS optical flow). Movers glow, stillness stays black.

## Heatmaps beat boxes

Box detectors collapse under overlap: F1 0.07 vs 0.71 point-based. Density maps cut counting error 265 to 105 MAE. (NWPU-Crowd)

## Pressure, not stillness

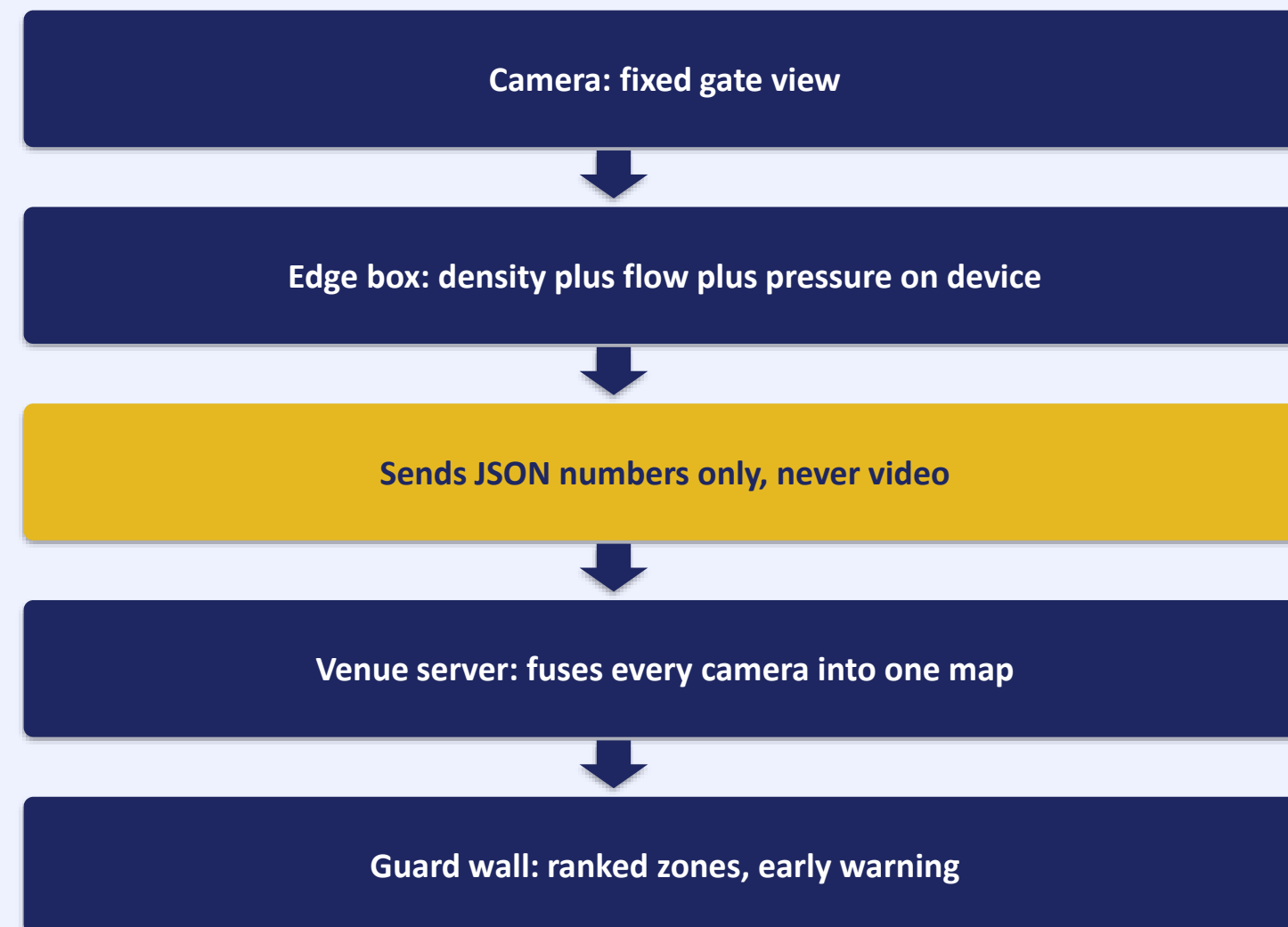
Pressure is density times uneven movement. Density or speed alone cannot warn. (Johansson 2008)

## Warns before stillness

Stop-and-go waves and rising pressure fire the alert while feet still move. Shown on Hajj 2006 footage; multi-site trials are the pilot's job.

# FEASIBILITY AND VIABILITY

## One camera, one edge box



One Jetson-class box (about \$250, 15 watts) per 1 to 4 cameras. Reuses existing CCTV. Pilot covers night, rain, oblique angles.

## Video stays put

Each edge box keeps a 10-minute rolling buffer and sends small JSON numbers, never raw video. Only a confirmed Red alert exports its clip, signed, to the evidence pack.

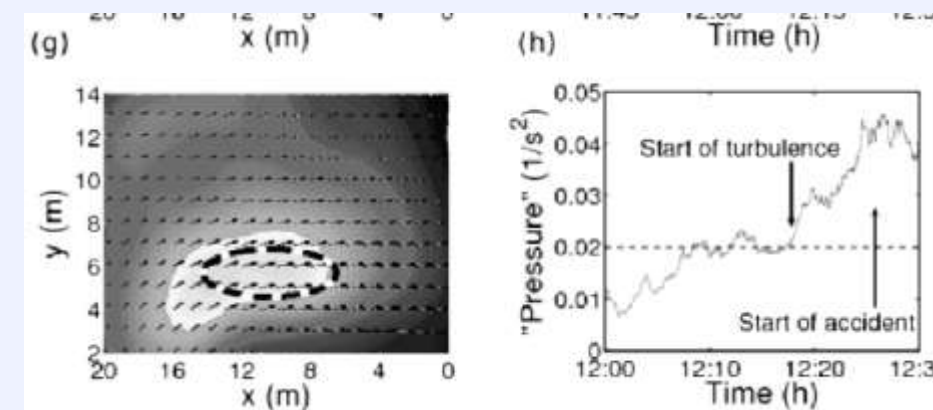
## Calibrated where it counts

Fixed gates are measured once; Kumbh ran 3-month surveys for its alarm. Open grounds and PTZ cameras run relative levels plus entry counts.

## Private by design

Heatmaps cannot show a face. Saved video is blurred, wiped after 7 days, zero identity in core. Police clips are access-logged under DPDP Act 2023 rules.

## Pressure warned first



Pressure  $p$  = density  $\times$  velocity variance. Hajj 2006: turbulence onset at 0.02 per s<sup>2</sup>, accident after. (Johansson et al.)

# IMPACT AND BENEFITS

## When pressure rises

### Yellow watch

Auto-pages the zone guard with an ack timer. Overflow gates get ready.

### Red alert

Guards get steps on their handsets and confirm. Unanswered reds climb the chain automatically.

### Handover

A sealed evidence pack goes to the police.

**Pilot targets: warn before critical, page guards in seconds, publish the false-alarm rate, log every step.**

### Before

A wall of screens, one tired guard, and risks get missed.

### After

Pressure at Gate B opens overflow at Gate A upstream. Every red has an owner, a timer, and a log.

### Organizers

Spot a growing jam early, and tell guards which zone to clear. Built for stations, mela grounds, stadiums.

### Police

Every alert, zone and action exports as one evidence pack in a click, with the signed 10-minute incident clip.

### Public

Faces stay blurred in saved video.



# RESEARCH AND REFERENCES

## The science behind it

### Count the overlap

[CSRNet, CVPR 2018](#)

[Bayesian Loss, ICCV 2019](#)

[DM-Count, NeurIPS 2020](#)

[P2PNet, ICCV 2021](#)

[STEERER / APGCC, 2023-24](#)

Tests: ShanghaiTech, QNRF, NWPU

Status: idea stage, side-track demo built. Build from: [DM-Count code \(MIT\)](#) [CSRNet model](#) [homography calibrator](#) [DeepStream custom-model docs](#)

### Ground and currents

[Chan perspective maps, CVPR 2008](#)

[Zhang ground-plane fusion, CVPR 2019](#)

[PACNN perspective, CVPR 2019](#)

[Mehran social force, CVPR 2009](#)

[Mehran streaklines, ECCV 2010](#)

[DIS optical flow \(OpenCV\)](#)

### Pressure and risk

[Helbing social force, 1995](#)

[Helbing crowd disasters, 2007](#)

[Johansson pressure, 2008](#)

[Hughes continuum, 2002](#)

[Nakayama divergence, BMVC 2025](#)

[Harding early warning, PLoS ONE 2011](#)

## Disasters, limits, footage

### Itaewon and Love Parade

159 dead, Seoul 2022, ignored calls. 21 dead, Duisburg 2010. Turbulence, not panic.

### Kumbh and Hathras

30 dead, Jan 2025, with AI live. 121 dead, Jul 2024, 80k permit vs 250k crowd.

### Danger thresholds

Fluid at 7 per m2. Serious above 4. Flow peaks at 2 to 3. (Fruin, Still)

### Test footage

Public clips: station, crossing, temple.  
[Wikimedia Commons](#)